

Physik II — FS 2024 — Prof. R. Wallny — Serie 9

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Exercise 1: Four-velocity

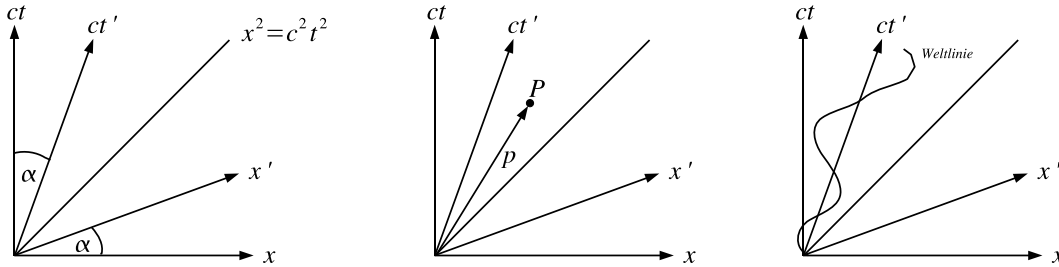
Objective: Four-velocity, computing four-vectors, reference frames

- a) Give the four-velocity of a particle with mass $m > 0$ in its restframe $\mathbf{K}$.
- b) Compute the four-velocity of this particle in a reference frame $\mathbf{K}'$, that is travelling with a constant velocity $\vec{v}$ with respect to $\mathbf{K}$.
- c) A second particle travels with a velocity $\vec{u}$ in the reference frame $\mathbf{K}$. Its speed is parallel to $\vec{v}$. Show that the norm of the corresponding four-velocity is equal in $\mathbf{K}$ and $\mathbf{K}'$.

Exercise 2: Minkowski diagrams

Objective: Four-vectors, Lorentz transformations, und Minkowski diagrams.

Many problems of special relativity, particularly problems that are constrained to one dimension, can be illustrated using so-called *Minkowski diagrams*. Such a diagram describes a system $\mathbf{K}'$ (with axes x' and ct') which moves at a velocity $u > 0$ with respect to a reference system $\mathbf{K}$ (with axes x and ct) by rotation of the spatial and of the temporal axes by an angle $\alpha = \arctan \beta$, where $\beta = u/c$ (see left figure).



A point P in space-time is referred to as an event, with the space-time vector p as its distance vector from the origin (see center figure). The squared four-length of the space-time vector p is defined through

$$p^2 = c^2 t^2 - x^2. \quad (1)$$

The term “world line” used to describe the path a given object has travelled in space-time (see right figure).

- a) Show that the angle between the axes of $\mathbf{K}$ and $\mathbf{K}'$ must be $\alpha = \arctan \beta$, and show that $\alpha < \pi/4$ for all $u < c$.
Hint: Express x' and ct' as a function of x and ct .

- b) Show that the four-length of the vector p is invariant under Lorentz boosts, i.e. show that $p'^2 = p^2$ (in the relativistic metric), where

$$p' = \Lambda p = \begin{pmatrix} \gamma & -\gamma\beta \\ -\gamma\beta & \gamma \end{pmatrix} \begin{pmatrix} ct \\ x \end{pmatrix}, \text{ with } \gamma = \frac{1}{\sqrt{1-\beta^2}}. \quad (2)$$

- c) Let the coordinates of an event P in system $\mathbf{K}$ be $P = (ct_0, x_0)$. Find the coordinates ct'_0, x'_0 of event P in system $\mathbf{K}'$. (Note that they can also be found graphically.) Draw the sets M and M' of events which occur simultaneously with P in $\mathbf{K}$ resp. $\mathbf{K}'$, i.e. the sets of events satisfying $ct = ct_0$ resp. $ct' = ct'_0$.
- d) Which functional form $ct = f(x, p)$ do all events Q have, that have the same four-distance $\sqrt{p^2}$ from the origin? Draw the set of such events (a rough sketch is sufficient).

Exercise 3: Causality

Goal: Apply Lorentz transformations and prove the concept of causality.

Two events 1 and 2 are separated by a difference in space $\Delta x = x_2 - x_1$ and time $\Delta t = t_2 - t_1$ in a reference frame $\mathbf{K}$.

- a) Use the Lorentz Transformation to show that for a reference frame $\mathbf{K}'$ moving at a speed of v along the x-axis the difference in time is described by

$$\Delta t' = t'_2 - t'_1 = \gamma(\Delta t - v \Delta x / c^2)$$

- b) In the reference frame $\mathbf{K}'$ the two events can only happen at the same time if Δx is greater than $c \Delta t$. Proof this statement.
- c) If one of the two events in $\mathbf{K}'$ *causes* the second event the distance Δx has to be smaller than $c \Delta t$ since a signal needs at least a time $\Delta x / c$ to travel from x_1 to x_2 in $\mathbf{K}$. Show in the case of $\Delta x < c \Delta t$ that $t'_2 > t'_1$ holds for all reference systems. This implies if the cause precedes the action in one system, it needs to precede it in all systems.
- d) Assume a signal can travel with $c' > c$. If in reference frame $\mathbf{K}$ the cause precedes the action by $\delta t = \Delta x / c'$ show that there exists a reference frame $\mathbf{K}'$ moving slower than the speed of light where the action precedes the cause.

This concludes the prove of footnote 4 in chapter 5.4.7.

Exercise 4: Electron-Positron annihilation

Objective: calculate with relativistic energy and momentum.

The positron is the electron's antiparticle: it has the same mass, but opposite charge. When a particle hits its antiparticle with low kinetic energy, the pair annihilates into electromagnetic energy in the form of photons (which are massless particles).

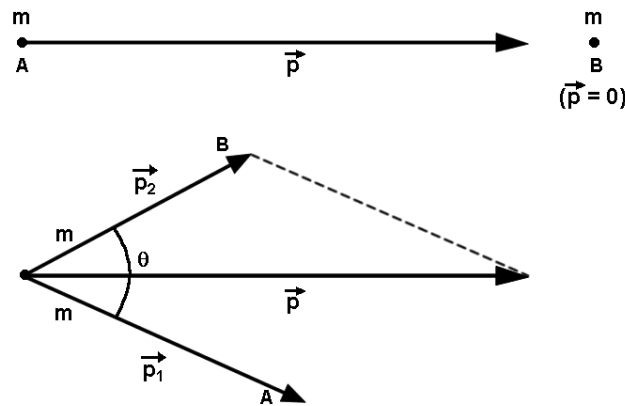
- a) Show that in such an annihilation, two or more photons are produced.

- b) An electron and positron are colliding head-on with a momentum $p = 1.55 \text{ GeV}/c$. The annihilation produces a new particle called J/Ψ . What is the total energy and speed of the electron and positrons? What is the mass of the produced particle?
Hint: The mass of the electron is about $0.5 \text{ MeV}/c^2$ and the J/Ψ is produced at rest.

Exercise 5: Relativistic collisions

Objective: Momentum and energy conservation, relativistic energy: rest and kinetic energy, ultrarelativistic vs. nonrelativistic case.

Let us consider a particle with mass m moving at a relativistic speed towards another particle which is at rest with the same mass (see figure). Before the collision, the incoming particle has a kinetic energy $E_{\text{kin}} = E_0$. After the collision, the two particles are moving at the same speed, thus they also have the same kinetic energy $E'_{\text{kin}} = E_0/2$. The angle between their trajectories is called θ .



- a) Using energy and momentum conservation, show that $\cos \theta = E_0 / (E_0 + 4mc^2)$.
- b) How large is this angle in the ultrarelativistic case ($E_0 \gg mc^2$) and in the nonrelativistic case ($E_0 \ll mc^2$)?